

Graph Anomaly Detection via Multiscale Contrastive Self-Supervised Learning From Local to Global

Xiaofeng Wang , Shuaiming Lai , Shuailei Zhu , Yuntao Chen , Laishui Lv , and Yuanyuan Qi 

Abstract—Graph anomaly detection is a challenging task in graph data mining, aiming to recognize unconventional patterns within a network. Recently, there has been increasing attention on graph anomaly detection based on contrastive learning due to its high adaptability to the sample imbalance problem. However, most existing work typically focuses on the contrast of local views while neglecting global comparison information, leading to suboptimal performance. To address this issue, we introduce a new multiscale contrastive self-supervised learning framework for graph anomaly detection (GADMCLG). Our approach incorporates local-level contrasts involving node–node and node–subgraph contrast, and global-level subgraph–subgraph contrast. The former mines localized abnormal information, while the latter is intended to capture global anomalous patterns. Specifically, our proposed subgraph–subgraph contrast adopts the h -order neighbor subgraph sampling instead of augmented subgraphs through edge perturbation. This sampling strategy ensures a comprehensive observation of the neighborhood surrounding the target node, thereby mitigating the introduction of extraneous noise and providing interpretability for the detected results. Furthermore, we incorporate a subgraph centralization technique to reduce the bias caused by the absolute position of subgraphs in the attribute space, which enhances the model’s ability to identify anomalies at different scales. Extensive experimental results on six real-world datasets demonstrate the effectiveness of our method and its superiority compared with state-of-the-art approaches.

Index Terms—Contrastive learning, graph anomaly detection, multiscale framework, subgraph centralization, subgraph sampling.

I. INTRODUCTION

GRAPH anomaly detection is one of the most important tasks in graph machine learning, which seeks to identify atypical graph patterns, including anomalous nodes, edges, and subgraphs in a given graph [1]. Node anomaly detection focuses on exploring individual data instances within the graph that deviate from expected patterns, making it the most prevalent task

in graph anomaly detection. Recently, graphs have emerged as a universal data structure for modeling diverse complex systems. Consequently, anomaly detection on graphs has garnered growing attention [2]. Graph anomaly detection techniques have found application in diverse fields, including fraud detection [3], social network analysis [4], recommendation systems [5], and network security [6]. In these domains, graph anomaly detection plays a crucial role in informing decision-making processes and facilitating effective problem-solving endeavors.

Traditional anomaly detection approaches face challenges in effectively addressing this issue due to the intricate nature of on-Euclidean graph-structured data [1]. Specifically, graphs emphasize the complex structural relationships between objects, which provides valuable auxiliary information for anomaly detection. In online social networks, malicious social bots usually build relationships with a large number of normal users to obtain user privacy information and can also disguise themselves by imitating the behavior of normal users [7]. In such cases, traditional techniques may not be able to distinguish between fake users and normal users based solely on attribute information. Furthermore, the differences in topology and attributes give rise to two principal categories of graph anomalies: structural anomalies and attribute anomalies [8] in Fig. 1. Nodes A and C are identified as attributed anomalies, while nodes A and B are identified as structural anomalies. The former refers to nodes within the graph that exhibit dissimilar interaction behaviors in comparison with other nodes, signifying disparities in the structural connectivity of the network. The latter involves interconnected nodes that exhibit unusual node attributes, highlighting the inconsistency of node attributes.

Over the past decade, numerous graph anomaly detection methods have been proposed. Early work such as LOF [9], AMEN [10], and ANOMALOUS [11] primarily utilized specific domain knowledge on graphs to identify anomalies, relying on manual feature engineering and statistical models. These conventional approaches fail to capture the nonlinear characteristics of nodes and scale to large-scale graphs. To address these limitations, deep learning has been introduced to uncover anomalous graph patterns, leveraging its success in representation learning and pattern recognition [12]. Deep neural networks with nonlinearity exhibit significant advantages in handling large-scale data and learning distributed representations for uncovering anomalies [13]. To capture more

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The authors are with the School of Information Engineering, China Jiliang University, Hangzhou 310018, China (e-mail: xfwang@cjljlu.edu.cn).

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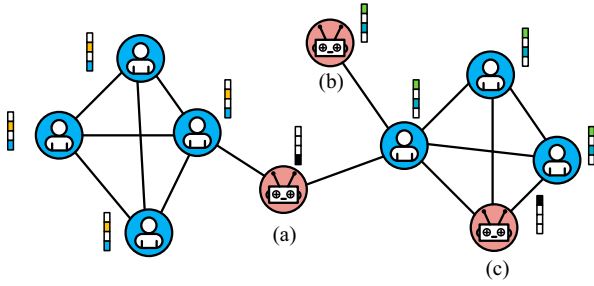


Fig. 1. Toy example of two basic types of graph anomalies.

discriminated information in graph data for anomaly detection, graph representation learning techniques have been widely exploited [14], [15], [16]. For example, Hu et al. [17] proposed an effective embedding approach to identify structural anomalies by analyzing the relationship between nodes and communities. Moreover, many typical shallow models for graph embedding such as DeepWalk [18] and node2vec [19] have been used for generating node representations in graph anomaly detection.

Recently, graph neural networks (GNNs) have been introduced to graph anomaly detection, attracting increasing attention due to their highly expressive capability in graph representation learning [8]. As a pioneering work, DOMINANT [20] utilizes a graph auto-encoder (GAE) based on a graph convolutional network (GCN) to identify anomalies in an unsupervised setting, where anomaly scores for nodes are calculated by leveraging the reconstruction errors of node attributes and graph structure. Subsequently, some variants were proposed, such as SpecAE [21], ALAMR [22], and ComGA [23]. GAE-based methods do not directly identify anomalous targets, making it difficult to fully utilize the rich information present in the attribute network [8]. Moreover, based on the self-supervised learning paradigm, CoLA [24] introduces a novel approach by combining contrastive learning with GNNs, which identifies anomalies by measuring the consistency between a target node and its neighboring nodes. On this basis, ANEMONE [25] captures more subtle abnormal information by building additional contrast between the target node and its masked nodes. Furthermore, GCAD [26] identifies anomalies by comparing the similarity between subgraphs. However, these methods typically employ a local contrastive mode or a single-scale contrast, while ignoring potential global anomaly information in complex graph-structured data.

To address this issue, we propose a novel graph anomaly detection method via multiscale contrastive self-supervised learning from local to global, denoted as GADMCLG. It integrates the local- and global-level contrasts to comprehensively detect anomalies in attributed graphs. Specifically, the local-level contrasts include node–node and node–subgraph contrasts. We utilize the node–subgraph contrast to capture potential matching patterns between each normal node and its corresponding neighbor subgraph sampled through random walk (RW). Moreover, the node–node contrast is used to mine more subtle local abnormal information. For the global-level contrast, we propose a subgraph–subgraph contrast based on the

h -order neighbor subgraph sampling strategy. Different from the existing method [27] that relies on edge modification to construct subgraph–subgraph contrast, our approach ensures a comprehensive observation of the neighborhood of the target node from local to global perspectives, thereby mitigating the introduction of extraneous noise. Furthermore, we perform subgraph centralization on all subgraphs to reduce the deviation caused by their absolute position in the attribute space. Finally, we design a multiscale contrastive learning framework by combining both local and global-level contrasts to identify potential abnormal nodes. The main contributions are summarized as follows.

- 1) We propose a novel graph anomaly detection method that leverages the synergy between local and global contrastive patterns to effectively identify node anomalies in attributed graphs.
- 2) We investigate the effects of various subgraph sampling methods on contrastive learning and conclude that the h -order neighborhood subgraph sampling strategy is the most suitable for the graph anomaly detection task.
- 3) Extensive experiments on six real-world datasets demonstrate the effectiveness of the proposed method and its advantages compared to state-of-the-art baselines.

The rest of the sections of this article are organized as follows. In Section II, we provide a general review of related work. Section III introduces fundamental concepts employed in this work and outlines the task objective of anomaly detection on attributed networks. We provide a detailed presentation of the proposed method in Section IV. Section V evaluates and analyzes the experimental results conducted across various datasets. Finally, Section VI presents the conclusion of our work.

II. RELATED WORK

Our work addresses the issue of graph anomaly detection using graph contrastive learning in attributed graphs. In this section, we first review the primary methods of graph anomaly detection and then introduce the pertinent work on graph contrastive learning.

A. Graph Anomaly Detection

Graph anomaly detection is a crucial topic in graph data mining, primarily employed to identify patterns that deviate from the majority within a graph. Much early work has been proposed to identify node anomalies. Breuning et al. [9] introduced a density-based local method LOF, which identifies anomalous nodes by comparing the attribute characteristics of their neighboring nodes. AMEN [10] focuses on identifying anomalous entities based on their local connectivity within the graph. Additionally, ANOMALOUS [11] utilizes a joint framework to leverage both structural properties and node attributes to reveal inherent patterns and identify anomalies. These traditional approaches primarily utilized specific domain knowledge on graphs to identify anomalies but failed to capture the nonlinear features of nodes, lacking deeper insight into graph data [1].

GNNs have emerged as an effective tool for graph representation learning, with the ability to extract complex nonlinear information. Consequently, numerous GNN-based methods have been proposed to uncover graph anomalies by learning expressive and informative node representations. DOMINANT [20] first introduces an anomaly detection approach based on GAE, which uncovers node anomalies by reconstructing both graph structure and node attributes in a supervised manner. SpecAE [21] utilizes GAE to extract low-dimensional embedding and performs anomaly detection via density estimation. Furthermore, ComGA [23] proposes a community-aware structure encoder and integrates a customized deep graph convolutional network to address the challenge of over-smoothing in GNNs. Moreover, AEGIS [28] extends GAE to the inductive setting, accommodating scenarios where previously unseen anomalies may be present. Despite the remarkable performance achieved by these methods in anomaly detection, it is noteworthy that they do not explicitly frame anomaly detection as a dedicated task, leading to suboptimal performance.

Recent studies have demonstrated that graph contrastive learning benefits graph-based machine learning tasks [29]. Consequently, the combination of graph contrastive learning for anomaly detection has received increasing attention. As a local contrastive learning method, CoLA [24] identifies anomalies by calculating the consistency between the target node and its neighboring nodes. Furthermore, SL-GAD [30] develops both generative attribute regression and contrastive learning framework to detect graph anomalies. ANEMONE [25] further combines the node–node contrast with the node–subgraph contrast and focuses on mining the local anomalous information. Moreover, GCAD [26] introduces a method based on global contrastive learning, which adopts subgraph–subgraph contrast to mine global anomaly information but ignores local anomaly patterns. GCCAD [31] designs a context-aware GNN encoder to learn both node and context representations, detecting abnormal information through the similarity between them.

B. Graph Contrastive Learning

As one of the most crucial paradigms in graph self-supervised learning, graph contrastive learning extracts supervised signals for downstream tasks without explicit labels by maximizing the mutual information (MI) between instances possessing similar semantic information [29]. Graph contrastive learning has gained popularity for its ability to capture complex relationships and structures in graph-structured data. Considering the scale differences in MI between instances, the existing work on graph contrastive learning can be mainly divided into two types: same- and cross-scale contrastive learning [32].

The same-scale contrast approaches for contrastive learning can be further divided into node- and graph-level contrasts. The underlying assumption of node-level contrastive learning is that nodes with similar contextual information should have similar representations. Early methods such as DeepWalk [18] and Node2Vec [19] utilize RW to extract contextual details surrounding target nodes in unattributed networks. Furthermore, GraphSAGE [33] further extends RW to attribute networks

and introduces a new graph convolutional network to inductively learn node embeddings. Different from these methods, recent work has shifted toward exploring more intricate semantic information by employing graph augmentation techniques. GRACE [34] leverages node feature masking and edge dropping to create two contrastive views, bringing closer representations of the same node in different views. Moreover, GCA [35] introduces the adaptive augmentation of graph-structured data based on the underlying graph properties, enriching the semantic information. On the other hand, graph-level methods utilize similar augmentation techniques and contrastive frameworks and employ readout functions on node representations to learn graph-level embeddings. JOAO [36] introduces a joint augmentation optimization strategy, cohesively optimizing augmentation selection and contrastive objectives. Besides, CSSL [37] is constructed upon the foundation of MoCo [38] and reduces model overfitting via graph-level contrastive learning.

The cross-scale approaches for contrastive learning emphasize the global comparison of views at different scales. As the pioneering work, DGI [39] proposes a patch-global contrast that contrasts node-level embeddings with graph-level representations to maximize the MI between these distinct scales of representation so as to aid the graph encoder in capturing both localized and global semantic information. Furthermore, HDGI [40] extends DGI to heterogeneous graphs by aggregating node representations on various meta paths to calculate the final node representation. Moreover, EGI [41] captures advanced transferable graph knowledge by forcing node features to maintain a consistent distribution with graph structure and then maximizes the MI between node embeddings and their surrounding ego-graphs. From the perspective of context-global contrasts, BiGI [42] learns the graph-level representation of the input graph by aggregating two types of node embeddings in a bipartite graph, as well as learning the local context representation of the target edge between two nodes over the sampled original graph. Besides, HTC [43] maximizes the MI between the representation of the entire graph and the contextual embedding calculated through aggregated sampling graphs.

III. PROBLEM DEFINITION

In this section, we introduce the task of graph anomaly detection in attributed networks and provide an overview of the notations used throughout this work. Table I summarizes the key notations used in this article.

Attributed Networks: For the given attributed network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ denotes the node set with N nodes, and $\mathcal{E}$ indicates the edge set with M edges. Moreover, the adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ and feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ represent the graph-structured information and the node-attribute information, respectively. It notes that $\mathbf{A}_{ij} = 1$ indicates that there is an edge between node v_i and node v_j , otherwise $\mathbf{A}_{ij} = 0$.

Graph Anomaly Detection: The task of graph anomaly detection is to learn an anomaly scoring function $f(\cdot)$, which computes the anomaly score for each node. The higher value of the score indicates a greater likelihood that the node is anomalous.

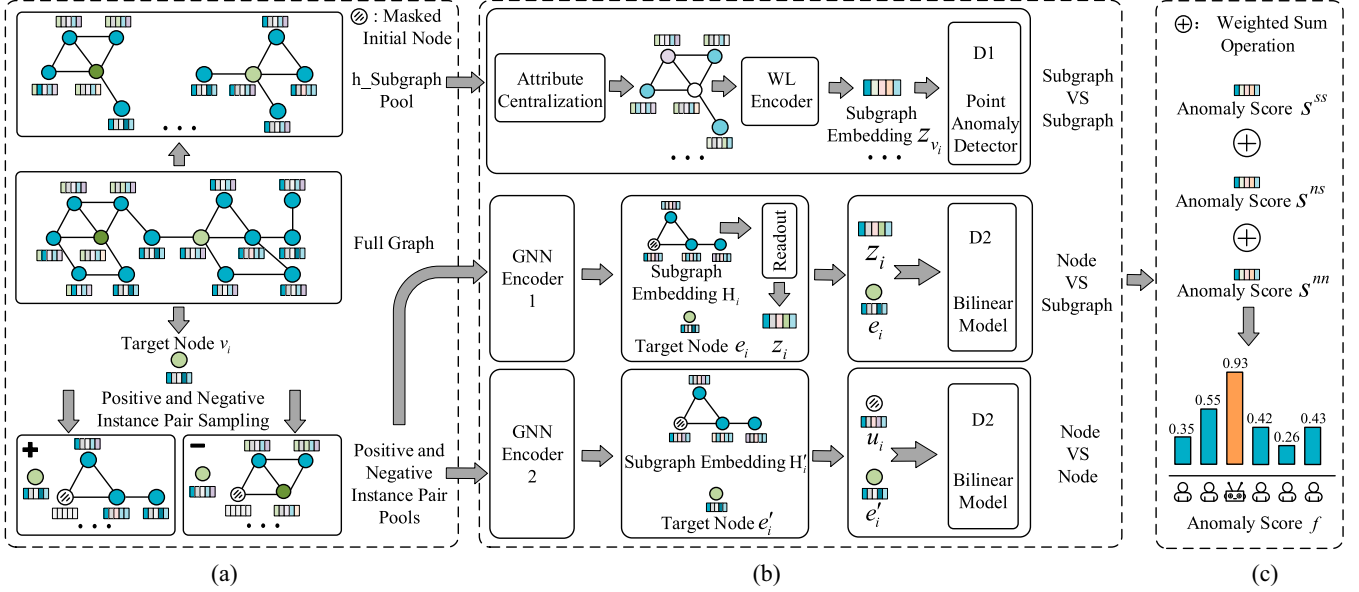


Fig. 2. Overview of the proposed GADMCLG. (a) Instance pair sampling module. (b) Multi-scale contrastive learning module. (c) Anomaly scoring module.

TABLE I
NOTATIONS AND DESCRIPTION TO GADMCLG

Notation	Description
$\mathcal{G}$	Attributed network.
$\mathcal{V}$	Node set of $\mathcal{G}$.
$\mathcal{E}$	Edge set of $\mathcal{G}$.
$\mathbf{A} \in \mathbb{R}^{n \times n}$	Adjacency matrix of $\mathcal{G}$.
$\mathbf{X} \in \mathbb{R}^{n \times d}$	Attribute matrix of $\mathcal{G}$.
$\mathbf{H}^{(\ell)} \in \mathbb{R}^{n \times d'}$	ℓ -th layer hidden embedding matrix.
$\mathbf{W}^{(\ell)} \in \mathbb{R}^{d' \times d'}$	ℓ -th layer network parameters.
$\phi(v_i)$	Final node embedding of node v_i .
Z_{v_i}	WL-embedded vector of the entire subgraph with Node v_i as the source node.
$f(v_i)$	Final anomaly score of node v_i .

IV. METHODOLOGY

In this section, we introduce the proposed method GADMCLG, and its overall framework is shown in Fig. 2. GADMCLG consists of three main components: an instance pair sampling module, a multiscale contrastive learning module, and an anomaly scoring module. Specifically, in the instance pair sampling module, we first adopt random walk with restart (RWR) [44] and h -order neighbor sampling strategy to construct two different subgraph sample pools for local- and global-level contrasts, respectively. Subsequently, the multiscale contrastive learning module extracts local anomalous information within local-level node-subgraph and node-node contrasts while simultaneously capturing global anomalous information through subgraph-subgraph contrast. Finally, the anomaly scoring module calculates the final anomaly score for each node in the graph, which is obtained through the weighted aggregation of anomaly scores derived from the three distinct contrastive mechanisms.

A. Contrastive Instance Pair Sampling

To comprehensively capture local and global anomalous information, GADMCLG employs contrastive learning across various scales. In the context of graph contrastive learning, contrastive instance pair sampling involves selecting pairs of nodes or subgraphs in a graph, where positive pairs might be nodes that are close in proximity or share similar attributes, and negative pairs are nodes that are far apart or have distinct features. It notes that the effectiveness of contrastive learning methods heavily depends on the definition of the contrastive instance pairs. In this article, we adopt the RWR and h -order neighbor sampling strategy instead of random sampling by edge perturbation to construct two different subgraph sample pools for local and global contrasts, respectively.

First, we define the node-subgraph contrast and node-node contrast to capture the local distribution pattern of nodes within the graph. Specifically, for the node-subgraph contrast, a target node needs to be selected first, and then the contrastive subgraphs of the target node are sampled by RWR and form contrastive positive pairs with the target node. In particular, to prevent the information leakage of the initial node of RWR during the contrastive learning process, we substitute the attribute vector with a zero vector to mask its node attribute. Subsequently, the node-node contrast is composed of the target node and masked initial node of RWR, which can mine more potential local anomalous information. It notes that positive and negative instance sampling strategies need to be adopted in local-level contrasts. For positive instance pairs, the initial node of the RWR corresponds to the target node, while for negative instance pairs, the initial node is randomly selected from all nodes except the target node.

Moreover, to establish the global connection between all the subgraphs, we define the subgraph-subgraph contrast that compares the structural similarity between different subgraphs

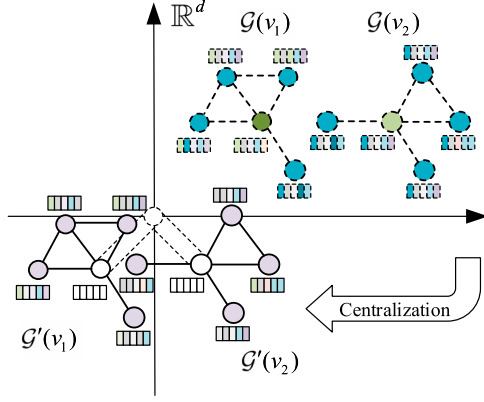


Fig. 3. Process of subgraph centralization.

globally. Different from the approach of subgraph sampling for the local contrasts, for subgraph–subgraph contrast, the subgraph of the target node is obtained by the h -order neighbor sampling strategy. Specifically, this sampling strategy extracts the adjacent nodes directly connected to the target node at a specified depth h .

B. Subgraph Centralization

As previously mentioned, subgraph–subgraph comparison involves comparing the structures of individual h -subgraphs. These structures are determined by the relative positions of nodes in the h -subgraph but are independent of the absolute positions of nodes in the node vector space. It notes that the absolute positions of subgraphs in the attribute space may cause bias in the similarity calculation between subgraphs. To address this issue, we perform attribute centralization on all subgraphs to generate the relative positions of all subgraphs. Specifically, we map all target nodes to the origin of the node-attribute space, and accordingly, other nodes in the corresponding subgraph are transformed to the same degree as their target nodes being transformed to the origin. This process is described in Fig. 3, where subgraphs $G'(v_1)$ and $G'(v_2)$ represent the state of subgraphs $G(v_1)$ and $G(v_2)$ after subgraph centralization.

C. Multiscale Contrastive Learning Module

After the instance pair sampling process, a multiscale contrastive learning module is used to capture robust and discriminative feature representations in attributed graphs for anomaly detection in a self-supervised manner. Contrastive learning has proven to be effective for graph anomaly detection. However, most approaches rely on local- or single-scale contrast, which may limit their capability to capture comprehensive anomalous information. In this work, we construct a multiscale graph contrastive learning module, including local- and global-level contrast. Specifically, this module consists of three components: node–subgraph contrast, node–node contrast, and global subgraph–subgraph contrast.

1) *Node–Subgraph Contrast*: To capture local anomalies, we first utilize the node–subgraph contrast to capture potential

matching patterns between each normal node and its corresponding neighbor subgraph sampled through RWR. Node–subgraph contrast follows the assumption that normal nodes should maintain a high degree of consistency with their neighboring nodes, while abnormal nodes should be the opposite. To calculate the consistency between the target node and its contrastive subgraph, we initially utilize a GNN encoder with a GCN layer [GNN encoder 1 in Fig. 2(b)] to map node attributes in the subgraph to the embedding space. The subgraph hidden-layer representation can be expressed as

$$\mathbf{H}_i^{(\ell+1)} = \sigma \left(\tilde{\mathbf{D}}_i^{-\frac{1}{2}} \tilde{\mathbf{A}}_i \tilde{\mathbf{D}}_i^{-\frac{1}{2}} \mathbf{H}_i^{(\ell)} \mathbf{W}^{(\ell)} \right) \quad (1)$$

where $\mathbf{H}_i^{(\ell)}$ denotes ℓ th layer hidden representation of i th subgraph, $\tilde{\mathbf{D}}_i^{-(1/2)} \tilde{\mathbf{A}}_i \tilde{\mathbf{D}}_i^{-(1/2)}$ is the normalized adjacency matrix of i th subgraph, $\mathbf{W}^{(\ell)}$ indicates the trainable weight parameters in ℓ th layer network, and $\sigma(\cdot)$ is the nonlinear activation function such as ReLU. We utilize a *Readout* function to calculate the final representation z_i of i th subgraph. The *Readout* function can be denoted as follows:

$$z_i = \text{Readout}(\mathbf{H}_i) = \sum_{j=1}^{n_i} \frac{(\mathbf{H}_i)_j}{n_i} \quad (2)$$

where $\mathbf{H}_i$ is the embedding of the i th subgraph after GCN mapping, and n_i is the number of all nodes in the i th subgraph.

Since the target node v_i lacks structural information, we utilize a multilayer perceptron (MLP) to obtain its embedding so that it can be contrasted with its contrastive subgraph in the same attribute space. The MLP hidden layer is defined as

$$h_i^{(\ell+1)} = \sigma(h_i^{(\ell)} \mathbf{W}^{(\ell)}) \quad (3)$$

where $\mathbf{W}^{(\ell)}$ is the weight matrix shared with GCN layer. e_i is the final embedding of the target node. After that, we adopt a *bilinear* model to calculate the similarity s_i of the target node and its contrastive subgraph

$$s_i = \text{Bilinear}(z_i, e_i) = \sigma(z_i \mathbf{W} e_i^\top). \quad (4)$$

It notes that the target node and its contrastive subgraph tend to be similar in positive pairs, while in negative pairs, they should be dissimilar. Therefore, based on this observation, we employ binary cross entropy (BCE) loss to train the contrast

$$\mathcal{L}_{NS} = - \sum_{i=1}^N (y_i \log(s_i) + (1 - y_i) \log(1 - s_i)) \quad (5)$$

where the value of y_i is 1 in positive pairs and 0 in negative pairs.

2) *Node–Node Contrast*: Similar to the node–subgraph contrast, node–node contrast focuses on capturing the local anomalous information at the node level. It notes that the representation of the masked initial node of RWR is aggregated from the other nodes present in the subgraph. We establish a positive pair between the target node and its masked node while forming a negative pair between the target node and other masked nodes that are not target nodes. After that, an additional GNN model

with a GCN layer [GNN encoder 2 in Fig. 2(b)] is employed to learn the representation of the subgraph

$$\mathbf{H}_i^{(\ell+1)} = \sigma \left(\tilde{\mathbf{D}}_i'^{-\frac{1}{2}} \tilde{\mathbf{A}}_i' \tilde{\mathbf{D}}_i'^{-\frac{1}{2}} \mathbf{H}_i^{(\ell)} \mathbf{W}'^{(\ell)} \right) \quad (6)$$

where $\mathbf{H}_i^{(\ell)}$ denotes ℓ th layer hidden representation of i th subgraph, $\tilde{\mathbf{D}}_i'^{-(1/2)} \tilde{\mathbf{A}}_i' \tilde{\mathbf{D}}_i'^{-(1/2)}$ is the normalized adjacency matrix of i th subgraph, and $\mathbf{W}'^{(\ell)}$ indicates the parameter matrix. $h_i'^{(\ell+1)} = \mathbf{H}_i^{(\ell+1)}[\mathbf{1}, :]$ is the representation of the masked initial node v_i' of RWR in the $(\ell + 1)$ th hidden layer and u_i is the final embedding of v_i' . We leverage MLP to transform the target node attribute to the same embedding space as the masked target node

$$h_i''^{(\ell+1)} = \sigma \left(h_i'^{(\ell)} \mathbf{W}'^{(\ell)} \right) \quad (7)$$

where $\mathbf{W}'^{(\ell)}$ is the same parameter matrix in Eq. (6). After MLP e_i' is the final embedding of the target node.

Similarly, we utilize a *Bilinear* model to calculate the similarity s_i' between e_i' and u_i . Subsequently, the loss associated with node–node contrast can be computed as follows:

$$\mathcal{L}_{NN} = - \sum_{i=1}^N (y_i' \log(s_i') + (1 - y_i') \log(1 - s_i')) \quad (8)$$

where the value of y_i' is 1 in positive pairs and is 0 in negative pairs.

3) *Subgraph–Subgraph Contrast*: Intuitively, each node in the network is more susceptible to the influence of its directly connected neighboring nodes. Therefore, abnormal nodes can often be represented through abnormal subgraphs. To capture global-level abnormal nodes, we score every node by calculating the similarity between all n subgraphs. The node v_i will have a higher anomaly score if its subgraph with v_i as the source node is different from most other subgraphs in the network.

Before calculating the similarity between subgraphs, we use *Weisfeiler–Lehman* (WL) [45] to learn subgraph embeddings. Specifically, we use the average neighborhood information of nodes to iteratively update their vectors. The iterative updating process of node v_i can be expressed as

$$\mathbf{X}^k(v_i) = \frac{1}{2}(\mathbf{X}^{k-1}(v_i) + \frac{1}{\deg(v_i)} \sum_{w \in \mathcal{N}(v_i)} \mathbf{X}^{k-1}(w)) \quad (9)$$

where $\mathbf{X}^k(v_i)$ denotes the vector of the node v_i in the k th iteration, $\deg(v_i)$ indicates the degree of node v_i , and $\mathcal{N}(v_i)$ is the set of one-hop neighbors of node v_i in the target subgraph. It notes that the final node embedding $\phi(v_i)$ of node v_i is the concatenation of vector from each iteration

$$\phi(v_i) = [\mathbf{X}^0(v_i), \dots, \mathbf{X}^k(v_i)]^T. \quad (10)$$

Therefore, the WL-embedded vector z_{v_i} of the entire target subgraph with the node v_i as the source node can be calculated as follows:

$$z_{v_i} = \frac{1}{|\mathbf{V}_{\text{sub-}v_i}|} \sum_{u \in \mathbf{V}_{\text{sub-}v_i}} \phi(u) \quad (11)$$

where $\mathbf{V}_{\text{sub-}v_i}$ indicates the subgraph with the node v_i as the source node.

D. Anomaly Scoring Module

After completing the training process in the multiscale contrastive learning module, the anomaly scoring module is used to calculate the node's anomaly score. For different contrasts, normal nodes typically exhibit obvious similarities or dissimilarities when compared to instances in positive and negative pairs. However, abnormal nodes are not distinguishable from positively and negatively aligned instances. Here, we compute the final anomaly score of each node from both local and global perspectives.

1) *Local Anomaly Score*: Local-level contrast consists of node–subgraph and node–node contrasts. A normal node exhibits similarity to its contrastive subgraph with it as the source node in positive pairs, while exhibiting dissimilarity in its negative pair. So that the anomaly score of the target node v_i can be calculated as follows:

$$\begin{aligned} s_i^{ns} &= s_i^n - s_i^p \\ s_i^{nn} &= s_i'^n - s_i'^p \\ s_i^{nns} &= \beta s_i^{ns} + (1 - \beta) s_i^{nn} \end{aligned} \quad (12)$$

where s_i^{ns} and s_i^{nn} denote the anomaly score to the node–subgraph and node–node contrast, respectively, s_i^n and s_i^p represent the similarity to the node–subgraph contrast in negative and positive pairs, respectively, $s_i'^n$ and $s_i'^p$ represent the similarity to the node–node contrast in negative and positive pairs, respectively, and β is the balance parameter of node–subgraph and node–node contrast. s_i^{nns} is the anomaly score of the local contrast after balancing the scores of node–subgraph and node–node contrast. It notes that the subgraph obtained from one RWR will lead to an incomplete observation of anomalous information. To this end, we conduct multiround detections, and the anomaly score of the target node v_i can be represented as follows:

$$\begin{aligned} \bar{s}_i &= \frac{1}{R} \sum_{r=1}^R s_i^{nns(r)} \\ s_i^{nns} &= \bar{s}_i + \sqrt{\frac{1}{R} \sum_{r=1}^R (s_i^{nns(r)} - \bar{s}_i)^2} \end{aligned} \quad (13)$$

where $\bar{s}_i$ is the mean from multiround detections, and R is the total rounds of anomaly detection.

2) *Global Anomaly Score*: After the WL-embedded vectors of all the subgraphs are computed, a point anomaly detector based on Euclidean distance is trained from them and produces an anomaly score s'' for the node v_i , which is the source node of the i th subgraph. Given that the depth of subgraphs may be greater than 1, a depth-based weighted anomaly score is introduced for further improvement of anomaly detection. Specifically, for the target node v_i , its anomaly score s_i^{ss} is the weighted score between all the source nodes and a larger weight will be given to s_j^{ss} if the node v_j is closer to node v_i

$$s_i^{ss} = \frac{\sum_{v_j \in \mathcal{V}} \lambda^{\ell(v_j, v_i)} s_j''}{\sum_{v_j \in \mathcal{V}} \lambda^{\ell(v_j, v_i)}} \quad (14)$$

where $\mathcal{V}$ is the set of all score nodes, $\lambda \in (0, 1)$ denotes the weight to anomaly score s_j'' , and $\ell(v_j, v_i)$ indicates the number of hops from node v_j to node v_i .

Finally, the anomaly score of the target node v_i can be denoted as follows:

$$f(v_i) = s_i^{nns} + \gamma s_i^{ss} \quad (15)$$

where γ is the balance parameter between the local anomaly score derived from node-subgraph and node-node contrasts and the global anomaly score from subgraph-subgraph contrast.

E. Loss Function

As shown in Fig. 2, the proposed GADMCLG model is divided into three phases: contrastive instance pair sampling, multiscale contrastive learning, and anomaly scoring. For the anomaly detection task, the entire model is first trained with sampled instance pairs in an unsupervised manner, and then the inference process is performed to obtain the anomaly score of each node. To combine the benefits of different contrasts, we optimize the joint loss function:

$$\mathcal{L} = \beta \mathcal{L}_{NS} + (1 - \beta) \mathcal{L}_{NN} \quad (16)$$

where $\beta \in (0, 1)$ is the balance parameter of node-subgraph and node-node contrast. It should be noted that, owing to the utilization of the WL propagation scheme in learning subgraph representations, the subgraph-subgraph contrast does not participate in joint training of node-subgraph and node-node contrasts. The overall procedures of our proposed GADMCLG are depicted in Algorithm 1.

V. EXPERIMENTS

In this section, we conduct extensive experiments on nine real-world graph datasets to demonstrate the excellent performance of GADMCLG. First, we provide an overview of the datasets and experimental setup. Then, we validate the effectiveness of the proposed method and evaluate the experimental results in a comparative view.

A. Datasets

We first estimate the performance of GADMCLG and its baseline methods on six injected anomaly graph datasets, which include two social network datasets (BlogCatalog and Flickr [46]) and five citation network datasets (ACM [47], Cora, Citeseer, Pubmed [48], and ogbn-arxiv [49]). Note that the ogbn-arxiv dataset, sourced from the open graph benchmark (OGB), is a large-scale graph dataset comprising over 169 000 nodes and 1.1 million edges. Since these datasets lack ground-truth anomalies, we completely followed the anomaly injection method in [11] and [14] to generate graph anomalies. Furthermore, to estimate the effectiveness of GADMCLG in real-world scenarios, we use two real-world anomaly graph datasets used in [23] and [50]. Among the real-world anomaly graph datasets, Amazon [51] represents a copurchase network in which nodes labeled as “amazonfail” are identified as abnormal, and Enron [52] is an email network where spammers are marked as abnormal nodes. Detailed information about these datasets is provided in Table II.

Algorithm 1 The Overall Procedures of GADMCLG.

Input: An attributed network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$; number of training epoches T ; batch size B ; subgraph depth h .
Output: Anomaly scoring function $f(\cdot)$.

- 1: **for** $t \in T$ **do**
- 2: Build positive and negative subgraph sample pools via RWR in $\mathcal{G}$.
- 3: Randomly divide $\mathcal{V}$ into batches of size B .
- 4: **for** $b \in \mathcal{B}$ **do**
- 5: In node-subgraph contrast, calculate the embedding similarity s_i of the target node v_i and the subgraph with v_i as the source node in positive and negative pairs via Eq. 4.
- 6: In node-node contrast, calculate the embedding similarity s_i' of the target node and the masked initial node of RWR in positive and negative pairs like node-subgraph contrast.
- 7: Calculate the joint loss L of node-subgraph and node-node contrasts via Eq. 16.
- 8: Backpropagation and update the trainable parameters of our model.
- 9: **end for**
- 10: **end for**
- 11: Calculate the joint anomaly score s^{nns} of node-subgraph and node-node contrasts via multi-round detections.
- 12: **for** $v \in \mathcal{V}$ **do**
- 13: Sample subgraphs with depth h .
- 14: Perform subgraph centralization on all subgraphs and Embed them to vectors via WL.
- 15: **end for**
- 16: Get the set Z_{all} of all subgraph vectors.
- 17: **for** $v \in \mathcal{V}$ **do**
- 18: Calculate anomaly score s_i'' for z_i with a point detector.
- 19: **end for**
- 20: Calculate the weighted scores s^{ss} via Eq. 14.
- 21: Calculate the final anomaly score f of all nodes.

TABLE II
DETAILS OF DATASETS

Dataset	Nodes	Edges	Features	Anomalies
BlogCatalog	5196	171 743	8189	300
Flickr	7575	239 738	12 407	450
ACM	16 484	71 980	8337	597
Cora	2708	5429	1433	150
Citeseer	3327	4732	3703	150
Pubmed	19 717	44 338	500	600
ogbn-arxiv	169 343	1 166 243	128	6000
Amazon	1418	3695	28	28
Enron	13 533	176 987	20	5

B. Experimental Settings

In this section, we describe the experimental settings, encompassing the baseline methods for comparative analysis, evaluation metrics, and parameter settings for our model.

1) *Baselines*: To verify the effectiveness and advantage of our proposed GADMCLG, we compare GADMCLG with six state-of-the-art baseline methods.

- a) *DOMINANT* [20]: DOMINANT is a typical method based on deep learning, which utilizes a GAE to calculate the reconstruction error of graph structure and node attributes, thereby detecting anomalies from both structural and attribute perspectives simultaneously.
- b) *CoLA* [24]: CoLA adopts a local contrastive self-supervised learning approach, leveraging a GNN-based encoder model to identify anomalies through an evaluation of the concordance between each node and its neighboring subgraph.
- c) *SL-GAD* [30]: SL-GAD integrates generative modeling and contrastive self-supervised learning in graph anomaly detection.
- d) *ComGA* [23]: ComGA introduces a community-aware structure encoder and integrates a customized deep graph convolutional network to address the challenge of over-smoothing in GNNs.
- e) *GCAD* [26]: GCAD is a global contrastive learning method for graph anomaly detection, calculating the similarity of subgraphs to dig global anomalous information.
- f) *GRADATE* [27]: It constructs a multiscale contrastive learning network frame with an augmented view.

2) *Metrics*: To evaluate the performance of GADMCLG and the baseline methods, we utilize the extensively adopted metric in anomaly detection, namely AUC-ROC. The ROC curve illustrates the graphical representation of the true positive rate (accurate identification of anomalous nodes) plotted against the false positive rate (misidentification of normal nodes as anomalies). Moreover, the AUC value, ranging from 0 to 1, quantifies the area under the ROC curve. A higher AUC value signifies better anomaly detection performance exhibited by the model under evaluation. Furthermore, the AUPR score serves as an extra evaluative metric for anomaly detection, particularly effective in contexts characterized by imbalanced data distributions. AUPR quantifies model performance by computing the area under the precision-recall curve across varying decision thresholds. Higher AUPR scores indicate superior model capability in correctly identifying positive samples.

3) *Parameter Settings*: For GADMCLG, the subgraph size is fixed to 4 for both node-subgraph contrast and node-node contrast. Both GNN models employ a single GCN layer and adopt ReLU as the activation function. Besides, the embedding dimension of both the target node and its contrastive subgraph is set to 64. During the training phase, the batch size $\mathcal{B}$, the learning rate α , and the total training epoch T for all datasets are set to 300, 0.001, and 400, respectively. During the inference phase, we conduct 256 round detections to ensure accurate detection results for each dataset. The balance parameter β between node-subgraph and node-node contrast is set to (0, 1). For subgraph-subgraph contrast, it adopts WL to embed all the subgraphs, and the depth h of the extracted subgraph for each dataset is set to 1. Moreover, the balance parameter γ between local contrasts and the global contrast is set to (0, 1).

To reproduce the results of other baseline methods, we adhere to the parameter settings specified in the respective publications. In cases where specific datasets do not have specified parameters, we employ a grid search to select the best parameters for optimal performance. Specifically, for DOMINANT, the balance parameters of attribute reconstruction error and structure reconstruction error on BlogCatalog, Flickr, ACM, Cora, Citeseer, and Pubmed datasets are set to 0.4, 0.7, 0.5, 0.4, 0.6, and 0.7, respectively. For CoLA, the subgraph size, embedding dimension, and sampling round are configured to 4, 64, and 256 for each dataset. In SL-GAD, the weight of the contrastive scores across all datasets is set to 1, with the weights of the generative scores being 0.6, 0.6, 0.7, 0.4, 0.5, and 0.4, respectively. For ComGA, the balance parameters of attribute reconstruction error and structure reconstruction error are set to 0.4, 0.4, 0.2, 0.2, 0.1, and 0.1, respectively. For GRADATE, the balance parameters of importance between the original view and the second view obtained through edge modification are 0.9, 0.5, 0.9, 0.9, 0.3, and 0.3, respectively. The balance parameters between subgraph- and node-level anomaly scores are 0.9, 0.3, 0.5, 0.3, 0.3, and 0.9. Last, the depth of subgraphs in GCAD for each dataset is set to 1.

C. Result Analysis

In this section, we evaluate the performance of GADMCLG in anomaly detection by comparing it with six state-of-the-art baselines. Fig. 4 illustrates the comparison of ROC curves. Moreover, the AUC and AUPR scores on the injected graph datasets and real-world anomaly graph datasets are presented in Tables III and IV, respectively. Based on the results, we make the following observations.

- 1) On the first six injected anomaly graph datasets, our proposed method GADMCLG outperforms all the baseline methods in terms of AUC. Specifically, GADMCLG obtains notable AUC gains of 1.67%, 0.77%, 4.24%, 3.80%, 2.46%, and 2.77% compared to the second-best performance on BlogCatalog, Flickr, ACM, Cora, Citeseer, and Pubmed, respectively. Simultaneously, the AUPR results demonstrate that GADMCLG significantly enhances positive sample detection relative to its baseline methods on most datasets, particularly achieving an approximate 23.13% increase in AUPR on the Citeseer dataset. These results suggest that GADMCLG exhibits superior overall performance and can effectively accommodate the varying requirements of different scenarios. The primary reason for the results is that our anomaly detection model based on multiscale contrastive learning can effectively capture both local and global anomaly information in the network simultaneously.
- 2) GADMCLG is successfully expanded to the large-scale network dataset ogbn-arxiv, overcoming a significant limitation of many baseline methods that are unable to perform anomaly detection due to extensive memory requirements. Although GADMCLG can also scale to even larger datasets, it necessitates increased computational time. This advantage is attributable to the

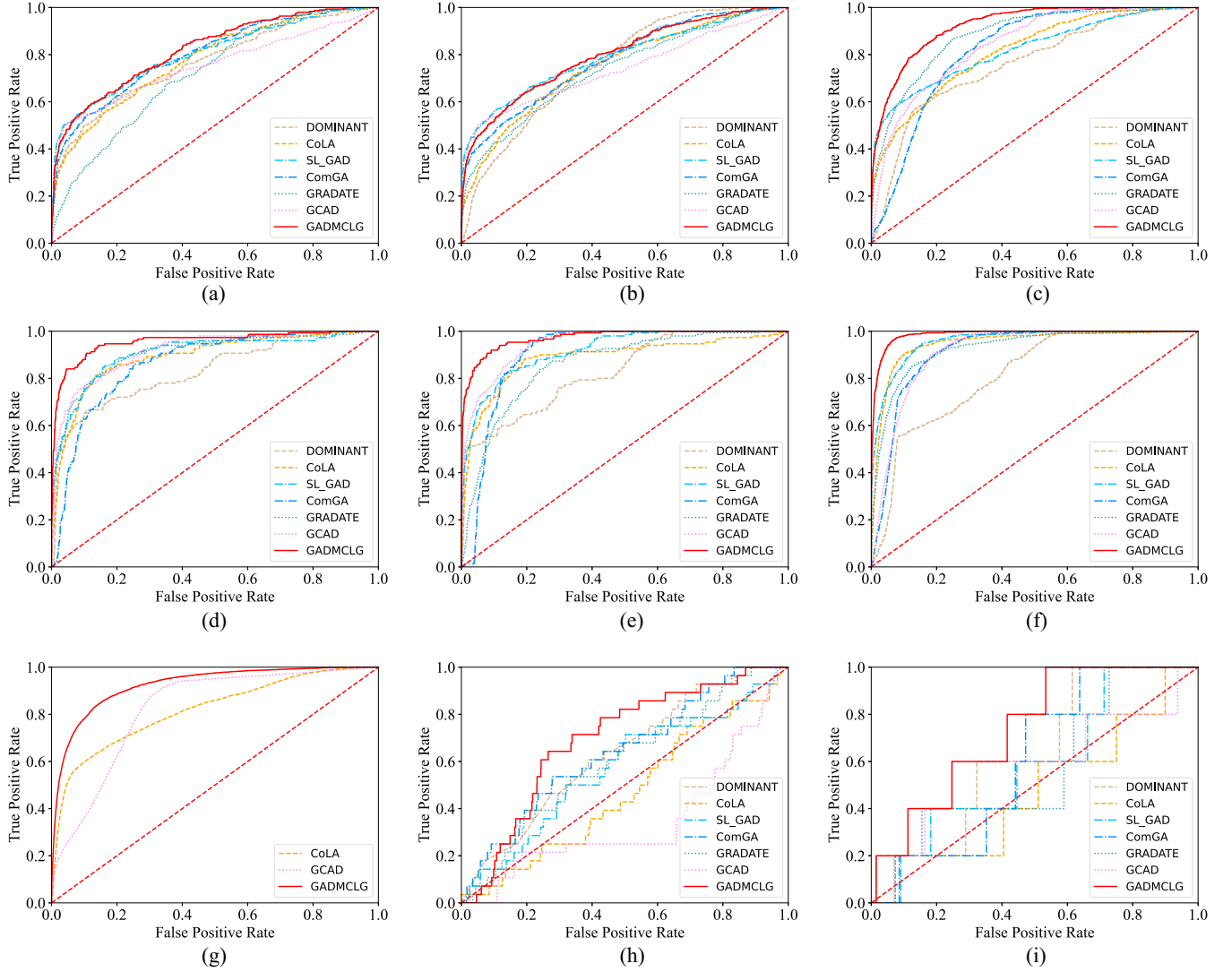


Fig. 4. ROC curves comparison on nine benchmark datasets. (a) BlogCatalog. (b) Flickr. (c) ACM. (d) Cora. (e) Citeseer. (f) Pubmed.

TABLE III
AUC AND AUPR VALUES COMPARISON ON THE INJECTED ANOMALY GRAPH DATASETS

methods	BlogCatalog		Flickr		ACM		Cora		Citeseer		Pubmed		ogbn-arxiv	
	AUC	AUPR	AUC	AUPR	AUC	AUPR	AUC	AUPR	AUC	AUPR	AUC	AUPR	AUC	AUPR
DOMINANT(2019)	0.7667	0.3071	0.7563	0.1534	0.7608	0.1036	0.8336	0.3354	0.8282	0.3500	0.8143	0.1023	OOM	OOM
CoLA(2021)	0.7780	0.3278	0.7522	0.2733	0.8151	0.3177	0.8911	0.4390	0.8877	0.4387	0.9495	0.4080	0.8171	<u>0.2318</u>
SL-GAD(2021)	<u>0.8047</u>	0.4387	0.7898	0.4463	0.8221	<u>0.4179</u>	0.9058	<u>0.5707</u>	0.9184	<u>0.5510</u>	0.9588	<u>0.6009</u>	OOM	OOM
ComGA(2022)	0.8008	0.3327	0.7768	0.3550	0.8210	0.1071	0.8643	0.2250	0.9054	<u>0.1921</u>	0.9159	0.1768	OOM	OOM
GRADATE(2023)	0.7190	0.1666	0.7370	0.2795	0.8817	0.3286	0.9075	0.5481	0.8601	0.2116	0.9279	0.3931	OOM	OOM
GCAD(2023)	0.7512	0.3630	0.7409	<u>0.4017</u>	0.8448	0.1853	0.9174	0.5323	<u>0.9475</u>	0.5345	0.9160	0.1813	<u>0.8223</u>	0.1712
GADMCLG(ours)	0.8214	<u>0.4036</u>	0.7975	0.3999	0.9241	0.4647	0.9554	0.7417	0.9721	0.7823	0.9865	0.7631	0.9238	0.4421

Note: The best performance on each dataset is highlighted in bold, and the second-best performance is marked underlined.

spatial complexity of GADMCLG being entirely independent of the number of nodes. Furthermore, GADMCLG shows significant performance advantages, with AUC and AUPR increased by 10.15% and 21.03%, respectively, compared to suboptimal performance.

3) As demonstrated in Table IV, GADMCLG achieves superior anomaly detection performance in terms of AUC on two real-world anomaly graph datasets, Amazon and Enron, with improvements of 3.1% and 9.88% over the suboptimal performance, respectively. Nevertheless, the

TABLE IV
AUC AND AUPR VALUES COMPARISON ON THE REAL-WORLD
ANOMALY GRAPH DATASETS

methods	Amazon		Enron	
	AUC	AUPR	AUC	AUPR
DOMINANT(2019)	0.6348	0.1168	<u>0.6363</u>	<u>0.0008</u>
CoLA(2021)	0.4705	0.0232	0.4721	0.0004
SL-GAD(2021)	0.5717	0.0251	0.5815	0.0005
ComGA(2022)	0.6401	0.0333	0.6021	0.0004
GRADATE(2023)	0.6003	0.0256	0.5671	0.0005
GCAD(2023)	<u>0.6510</u>	0.0146	0.5855	0.0005
GADMCLG(ours)	0.6820	<u>0.0452</u>	0.7351	0.0011

Note: The best performance on each dataset is highlighted in bold, and the second-best performance is marked underlined.

significant imbalance between real anomalies and normal nodes in the Amazon and Enron datasets poses substantial challenges for all methods in accurately identifying anomalies. This can be observed from the relatively low evaluation scores, especially in AUPR. GADMCLG achieved the highest AUPR score on Enron, but it performed lower than DOMINANT on Amazon. Addressing this limitation will be a key focus of our future research.

- 4) The GAE-based methods, DOMINANT and ComGA, cannot achieve satisfying results through network reconstruction error. The reason is that they do not directly consider anomaly detection as a specific task leading to suboptimal performance. In contrast, GADMCLG adopts a contrastive learning-based approach by assessing the consistency among nodes or subgraphs, thereby enabling the direct detection of graph anomalies and providing more reasonable explanations for detected anomalies.
- 5) The contrastive learning methods, CoLA, GCAD, and GADMCLG, exhibit improved performance compared to GAE-based methods. This suggests that contrastive learning methods more effectively identify anomalies by capturing the information consistency differences in attributes and structures within the network. Furthermore, compared to GADMCLG based on multiscale contrastive learning, the other two methods only utilize single-scale contrast, resulting in incomplete capture of anomalous information and, therefore, lowering performance.
- 6) Compared to another multiscale contrastive learning method GRADATE, GADMCLG achieved an average 7.11% performance improvement in terms of AUC metric. The reason is that GADMCLG adopts the h -order subgraph sampling strategy, while GRADATE uses edge modification to generate contrast pairs within subgraph-subgraph contrast. This type of graph augmentation has the potential to introduce additional noise interference, consequently impeding the efficacy of anomaly detection.

D. Parameter Analysis

In this section, we explore the hyperparameter sensitivity of two important balance parameters β and γ , and the performance of the proposed model. In this experiment, we investigate their

impacts on AUC. Both of these two hyperparameters are tuned within the range from 0 to 1 at intervals of 0.1. The experimental results on the six benchmark datasets are shown in Fig. 5.

As depicted in Fig. 5, the hyperparameters β and γ are found to be effective in enhancing the performance of the proposed model across all datasets. With the increase in parameter values, AUC has improved to varying degrees, especially on the Cite-seer and Pubmed datasets. The above experimental observations indicate that for BlogCatalog, Flickr, ACM, Cora, Citeseer, and Pubmed datasets, setting β to 0.2, 0.3, 0.3, 0.4, 0.8, and 0.6, respectively, alongside setting γ at 0.9, 0.2, 0.7, 0.8, 0.6, and 0.5, our proposed GADMCLG exhibits optimal performance for anomaly detection.

E. Ablation Study

In this section, we verify the effectiveness of the proposed multiscale contrastive strategy employed in GADMCLG and examine the influence of various subgraph sampling strategies on anomaly detection performance.

1) *Multiscale Contrastive Strategy*: To validate the effectiveness of our proposed GADMCLG based on multiscale contrastive learning, we perform an ablation study on the six datasets and evaluate the performance in terms of AUC. In this experiment, we define three variants: GADMCLG w/o ns, GADMCLG w/o nn, and GADMCLG w/o ss. Specifically, GADMCLG w/o ns represents the variant of GADMCLG without the node-subgraph contrast, GADMCLG w/o nn is its variant that excludes the node-node contrast, and GADMCLG w/o ss indicates only the node-subgraph and subgraph-subgraph contrasts are used for detecting anomalies. The performance variance results are presented in Table V.

The experimental results indicate an insightful observation that node-subgraph contrast manifests a more significant impact on the performance of anomaly detection in comparison with node-node contrast. This is because, for node-node contrast, we construct the comparison between the target node and the initial masked node of RWR. However, it notes that the representation of the masked initial node is generated by aggregating its neighboring attribute information. Consequently, node-node contrast essentially constitutes an alternative manifestation of node-subgraph contrast. Moreover, compared with GADMCLG w/o ss, GADMCLG achieves significant performance improvement because subgraph-subgraph contrast captures important global anomaly information in the network by calculating the anomaly score of their central nodes based on the similarity between subgraphs. Therefore, the multiscale contrastive strategy that combines local and global abnormal information leads to the best performance.

2) *Subgraph Centralization*: The calculation of subgraph similarity is a complex problem, and its accuracy is affected by the relative positions between subgraphs. To verify the impact of the subgraph centralization technique we used on the performance of graph anomaly detection, we remove this technique from the global comparison and define the variant as GADMCLG w/o sc. The experimental results on six datasets are shown in Fig. 6.

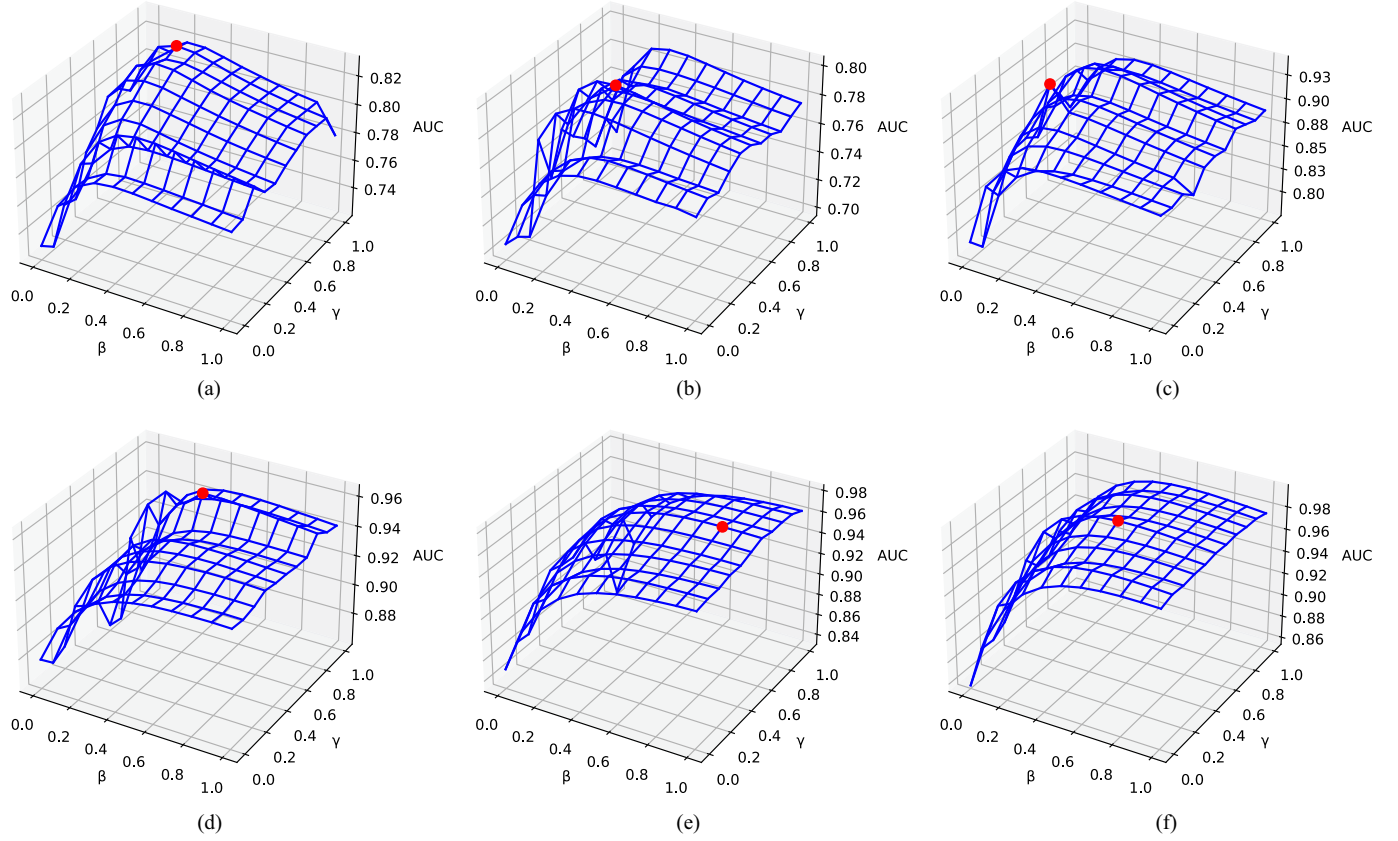


Fig. 5. Sensitivity analysis for the balance parameters β and γ w.r.t AUC. (a) BlogCatalog. (b) Flickr. (c) ACM. (d) Cora. (e) Citeseer. (f) Pubmed.

TABLE V
EFFECT OF DIFFERENT CONTRASTIVE STRATEGIES W.R.T AUC

Methods	BlogCatalog	Flickr	ACM	Cora	Citeseer	Pubmed
GADMCLG w/o ns	0.7697	0.7882	0.8789	0.9300	0.9601	0.9771
GADMCLG w/o nn	0.8118	0.7802	0.9077	0.9461	0.9631	0.9796
GADMCLG w/o ss	0.7976	0.7584	0.8730	0.9113	0.9105	0.9548
GADMCLG	0.8214	0.7975	0.9241	0.9554	0.9721	0.9865

Note: The best performance on each dataset is highlighted in bold.

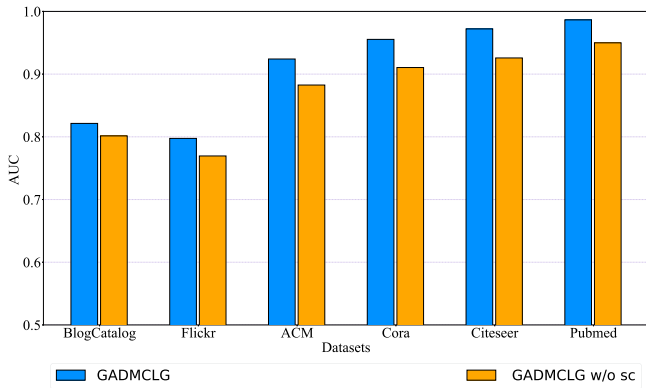


Fig. 6. Effect of subgraph centralization on the performance of GADMCLG.

Based on the above experimental results, we found that subgraph centralization greatly affects the final anomaly detection performance of our model. Without this important component,

GADMCLG experienced a significant performance decline on all six datasets. This indicates that subgraph centralization technology is very necessary for graph anomaly detection.

3) *Subgraph Sampling Strategy*: To investigate the impact of different subgraph sampling strategies on subgraph-subgraph contrast in graph anomaly detection, we compare the h-order sampling with two additional sampling strategies: RWR and intracommunity RW [53]. Correspondingly, we combine these two sampling strategies with multiscale contrastive learning and then define two variants GADMCLG-RWR and GADMCLG-CW, respectively. The comparative results are shown in Fig. 7.

It can be observed that the proposed GADMCLG performs best across all datasets compared to GADMCLG-RWR and GADMCLG-CW. This result demonstrates that the h-order neighbor subgraph sampling strategy is more suitable for subgraph-subgraph contrast than the other two sampling strategies. Further analysis reveals that RWR introduces a significant

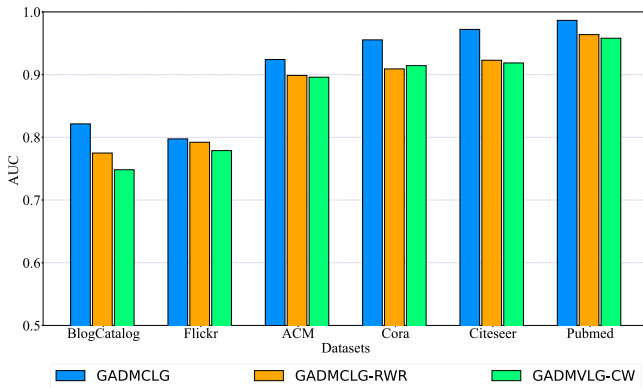


Fig. 7. Effect of different subgraph sampling strategies w.r.t. AUC.

degree of randomness, making it unable to fully capture the entire neighborhood observation related to the target node. Besides, intracommunity RW focuses on capturing the community structure of the target node, ignoring anomalous attribute information. However, the h -order neighbor subgraph sampling strategy simultaneously captures the subgraph structure and neighbor node-attribute information. Consequently, subgraph-subgraph contrast for graph anomaly detection depends on a comprehensive observation of the target node's neighborhood.

VI. CONCLUSION

In this article, we proposed a graph anomaly detection model via multiscale contrastive self-supervised learning, navigating from local to global contexts. To uncover underlying anomaly nodes in attributed networks, we incorporate local-level contrast and global-level contrast to enhance the model's capability to discern anomalous patterns across different scales, contributing to a more comprehensive and effective detection of anomalies within graph-structure data. Moreover, we introduce the subgraph centralization strategy to mitigate the influence of the absolute position of subgraphs in the attribute space on global subgraph contrast, thereby enhancing the detection accuracy of the proposed model. The effectiveness of the proposed approach has been validated on six real-world benchmark datasets, and experimental results demonstrate its superiority over competitive counterparts.

Furthermore, since subgraph contrast relies on global observations of the target node's neighborhood, the size of its neighbor subgraphs exhibits diversity. This diversity makes it challenging to employ subgraph embedding methods with the same contrast as the local contrasts, rendering them unable to participate in joint training. In the future, we will try to solve this problem. Moreover, the interpretability of the results constitutes a crucial concern in graph anomaly detection. We will extend graph contrastive learning to enhance the interpretability of graph anomaly detection models.

REFERENCES

- [1] X. Ma et al., "A comprehensive survey on graph anomaly detection with deep learning," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 12, pp. 12012–12038, Dec. 2023.
- [2] A. D. Pazho, G. A. Noghre, A. A. Purkayastha, J. Vempati, O. Martin, and H. Tabkhi, "A survey of graph-based deep learning for anomaly detection in distributed systems," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 1, pp. 1–20, Jan. 2024.
- [3] L. Zheng, G. Liu, C. Yan, and C. Jiang, "Transaction fraud detection based on total order relation and behavior diversity," *IEEE Trans. Comput. Soc. Syst.*, vol. 5, no. 3, pp. 796–806, Sep. 2018.
- [4] A. Dey, B. R. Kumar, B. Das, and A. K. Ghoshal, "Outlier detection in social networks leveraging community structure," *Inf. Sci.*, vol. 634, pp. 578–586, 2023.
- [5] J. Meira et al., "Anomaly detection on natural language processing to improve predictions on tourist preferences," *Electronics*, vol. 11, no. 5, p. 779, 2022.
- [6] B. Alwasel, A. Aldribi, M. Alreshodi, I. S. Alsukayti, and M. Al-suhaibani, "Leveraging graph-based representations to enhance machine learning performance in IIoT network security and attack detection," *Appl. Sci.*, vol. 13, no. 13, p. 7774, 2023.
- [7] Q. Guo, H. Xie, Y. Li, W. Ma, and C. Zhang, "Social bots detection via fusing BERT and graph convolutional networks," *Symmetry*, vol. 14, no. 1, p. 30, 2021.
- [8] H. Kim, B. S. Lee, W.-Y. Shin, and S. Lim, "Graph anomaly detection with graph neural networks: Current status and challenges," *IEEE Access*, vol. 10, pp. 111820–111829, 2022.
- [9] M. M. Breunig, H. P. Kriegel, R. T. Ng, and J. Sander, "LOF: Identifying density-based local outliers," in *Proc. ACM SIGMOD Int. Conf. Manag. Data*, 2000, pp. 93–104.
- [10] B. Perozzi and L. Akoglu, "Scalable anomaly ranking of attributed neighborhoods," in *Proc. 16th SIAM Int. Conf. Data Mining (SDM)*, 2016, pp. 207–215.
- [11] Z. Peng, M. Luo, J. Li, H. Liu, and Q. Zheng, "ANOMALOUS: A joint modeling approach for anomaly detection on attributed networks," in *Proc. 27th Int. Joint Conf. Artif. Intell. (IJCAI)*, Jul. 2018, pp. 3515–3519.
- [12] G. Pang, C. Shen, L. Cao, and A. V. D. Hengel, "Deep learning for anomaly detection: A review," *ACM Comput. Surv.*, vol. 54, no. 2, pp. 1–38, 2021.
- [13] S. Bandyopadhyay, L. N. S. V. Vivek, and M. N. Murty, "Outlier resistant unsupervised deep architectures for attributed network embedding," in *Proc. 13th Int. Conf. Web Search Data Mining*, Jan. 2020, pp. 25–33.
- [14] P. Cui, X. Wang, J. Pei, and W. Zhu, "A survey on network embedding," *IEEE Trans. Knowl. Data Eng.*, vol. 31, no. 5, pp. 833–852, May 2019.
- [15] J. Zhu, J. Wang, Y. Shan, S. Yu, G. Chen, and Q. Xuan, "DeepInsight: Topology changes assisting detection of adversarial samples on graphs," *IEEE Trans. Comput. Soc. Syst.*, vol. 11, no. 1, pp. 76–88, Feb. 2024.
- [16] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, Jan. 2021.
- [17] R. Hu, C. C. Aggarwal, S. Ma, and J. Huai, "An embedding approach to anomaly detection," in *Proc. IEEE Int. Conf. Data Eng. (ICDE)*, May 2016, pp. 385–396.
- [18] B. Perozzi, R. Al-Rfou, and S. Skiena, "DeepWalk: Online learning of social representations," in *Proc. 20th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, Aug. 2014, pp. 701–710.
- [19] A. Grover and J. Leskovec, "node2vec: Scalable feature learning for networks," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, Aug. 2016, pp. 855–864.
- [20] K. Ding, J. Li, R. Bhanushali, and H. Liu, "Deep anomaly detection on attributed networks," in *Proc. SIAM Int. Conf. Data Mining (SDM)*, May 2019, pp. 594–602.
- [21] Y. Li, X. Huang, J. Li, M. Du, and N. Zou, "SpecAE: Spectral autoencoder for anomaly detection in attributed networks," in *Proc. 28th Int. Conf. Inf. Knowl. Manage.*, Nov. 2019, pp. 2233–2236.
- [22] Z. Peng, M. Luo, J. Li, L. Xue, and Q. Zheng, "A deep multi-view framework for anomaly detection on attributed networks," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 6, pp. 2539–2552, Jun. 2022.
- [23] X. Luo et al., "ComGA: Community-aware attributed graph anomaly detection," in *Proc. 15th ACM Int. Conf. Web Search Data Mining (WSDM)*, Feb. 2022, pp. 657–665.
- [24] Y. Liu, Z. Li, S. Pan, C. Gong, C. Zhou, and G. Karypis, "Anomaly detection on attributed networks via contrastive self-supervised learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 6, pp. 2378–2392, Jun. 2022.
- [25] M. Jin, Y. Liu, Y. Zheng, L. Chi, Y. Li, and S. Pan, "ANEMONE: Graph anomaly detection with multi-scale contrastive learning," in *Proc. 30th Int. Conf. Inf. Knowl. Manage. (CIKM)*, Oct. 2021, pp. 3122–3126.
- [26] Z. Zhang, K. Ting, G. Pang, and S. Song, "Subgraph centralization: A necessary step for graph anomaly detection," in *Proc. SIAM Int. Conf. Data Mining (SDM)*, 2023, pp. 703–711.

- [27] J. Duan et al., “Graph anomaly detection via multi-scale contrastive learning networks with augmented view,” in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 6, Jun. 2023, pp. 7459–7467.
- [28] K. Ding, J. Li, N. Agarwal, and H. Liu, “Inductive anomaly detection on attributed networks,” in *Proc. 29th Int. Joint Conf. Artif. Intell.*, Jan. 2021, pp. 1288–1294.
- [29] Y. Xie, Z. Xu, J. Zhang, Z. Wang, and S. Ji, “Self-supervised learning of graph neural networks: A unified review,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2412–2429, Feb. 2023.
- [30] Y. Zheng, M. Jin, Y. Liu, L. Chi, K. T. Phan, and Y.-P. P. Chen, “Generative and contrastive self-supervised learning for graph anomaly detection,” *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 12, pp. 12220–12233, Dec. 2023.
- [31] B. Chen et al., “GCCAD: Graph contrastive coding for anomaly detection,” *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 8, pp. 8037–8051, Aug. 2023.
- [32] Y. Liu et al., “Graph self-supervised learning: A survey,” *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 6, pp. 5879–5900, Jun. 2023.
- [33] W. Hamilton, Z. Ying, and J. Leskovec, “Inductive representation learning on large graphs,” in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1025–1035.
- [34] Y. Zhu, Y. Xu, F. Yu, Q. Liu, S. Wu, and L. Wang, “Deep graph contrastive representation learning,” 2020, *arXiv:2006.04131*.
- [35] Y. Zhu, Y. Xu, F. Yu, Q. Liu, S. Wu, and L. Wang, “Graph contrastive learning with adaptive augmentation,” in *Proc. WWW*, Apr. 2021, pp. 2069–2080.
- [36] Y. You, T. Chen, Y. Shen, and Z. Wang, “Graph contrastive learning automated,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, Jul. 2021, pp. 12121–12132.
- [37] J. Zeng and P. Xie, “Contrastive self-supervised learning for graph classification,” in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, no. 12, May 2021, pp. 10824–10832.
- [38] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, “Momentum contrast for unsupervised visual representation learning,” in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 9729–9738.
- [39] P. Veličković, W. Fedus, W. L. Hamilton, P. Liò, Y. Bengio, and R. D. Hjelm, “Deep graph infomax,” 2018, *arXiv:1809.10341*.
- [40] Y. Ren, B. Liu, C. Huang, P. Dai, L. Bo, and J. Zhang, “Heterogeneous deep graph infomax,” 2019, *arXiv:1911.08538*.
- [41] Q. Zhu, C. Yang, Y. Xu, H. Wang, C. Zhang, and J. Han, “Transfer learning of graph neural networks with ego-graph information maximization,” in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 1766–1779.
- [42] J. Cao, X. Lin, S. Gao, L. Liu, T. Liu, and B. Wang, “Bipartite graph embedding via mutual information maximization,” in *Proc. WSDM*, Mar. 2021, pp. 635–643.
- [43] Z. Liu, C. Wang, C. Han, and T. Guo, “Learning graph representation by aggregating subgraphs via mutual information maximization,” *Neurocomputing*, vol. 518, 2023, Art. no. 126392.
- [44] H. Tong, C. Faloutsos, and J.-y. Pan, “Fast random walk with restart and its applications,” in *Proc. 6th IEEE Int. Conf. Data Mining (ICDM)*, Dec. 2006, pp. 613–622.
- [45] M. Togninalli, E. Ghisu, F. Llinares-López, B. Rieck, and K. Borgwardt, “Wasserstein Weisfeiler-Lehman graph kernels,” in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 6439–6449.
- [46] L. Tang and H. Liu, “Relational learning via latent social dimensions,” in *Proc. 15th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, Jun. 2009, pp. 817–826.
- [47] J. Tang, J. Zhang, L. Yao, J. Li, L. Zhang, and Z. Su, “ArnetMiner: Extraction and mining of academic social networks,” in *Proc. 14th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, Aug. 2008, pp. 990–998.
- [48] P. Sen, G. Namata, M. Bilgic, L. Getoor, B. Galligher, and T. Eliassi-Rad, “Collective classification in network data,” *AI Mag.*, vol. 29, no. 3, p. 93, 2008.
- [49] W. Hu et al., “Open graph benchmark: Datasets for machine learning on graphs,” in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 22118–22133, 2020.
- [50] Y. Pei, T. Huang, W. van Ipenburg, and M. Pechenizkiy, “ResGCN: Attention-based deep residual modeling for anomaly detection on attributed networks,” *Mach. Learn.*, vol. 111, no. 2, pp. 519–541, 2022.
- [51] E. Müller, P. I. Sánchez, Y. Mülle, and K. Böhm, “Ranking outlier nodes in subspaces of attributed graphs,” in *Proc. 29th ICDEW*, 2013, pp. 216–222.
- [52] V. Metsism, I. Androutsopoulos, and G. Paliouras, “Spam filtering with Naive Bayes—Which naive Bayes?” in *Proc. CEAS*, vol. 17, 2006, pp. 28–69.

- [53] P. Pham, L. T. Nguyen, B. Vo, and U. Yun, “Bot2Vec: A general approach of intra-community oriented representation learning for bot detection in different types of social networks,” *Inf. Syst.*, vol. 103, 2022, Art. no. 101771.



machine learning.



Xiaofeng Wang received the Ph.D. degree in information and communication engineering from Shanghai Jiao Tong University (SJTU), Shanghai, China, in 2017.

He is currently a Lecturer with the College of Information Engineering, China Jiliang University, Hangzhou, China. He served as a Postdoctoral Fellow in computer science with SJTU, and as a Visiting Scholar with Concordia University, Montreal, Canada. His research interests include network science, graph neural networks, data mining, and

Shuaiming Lai received the B.S. degree in electronic science and technology from Suzhou City University, Suzhou, China, in 2021. He is currently working toward the M.S. degree in artificial intelligence with China Jiliang University, Hangzhou, China.

His research interests include graph anomaly detection, graph neural networks, social network analysis, and machine learning.



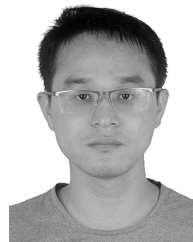
Shuailei Zhu received the B.S. degree in communication engineering in 2021 from the College of Modern Science and Technology, China Jiliang University, Hangzhou, China, where he is currently working toward the M.S. degree in communication engineering.

His research interests include community detection, graph neural networks, and machine learning.



Yuntao Chen received the B.S. degree in communication engineering in 2021 from the College of Modern Science and Technology, China Jiliang University, Hangzhou, China, where he is currently working toward the M.S. degree in communication engineering.

His research interests include signal graph classification, graph neural networks, and machine learning.



Laishui Lv received the Ph.D. degree in computer science and technology from Nanjing University of Science and Technology, Nanjing, China, in 2021.

He is currently a Lecturer with the College of Information Engineering, China Jiliang University, Hangzhou, China. His research interests include artificial intelligence, complex networks, statistical analysis, and nonlinear optimization.



Yuanyuan Qi received the Ph.D. degree in information and communication engineering from Beijing University of Posts and Telecommunications (BUPT), Beijing, China, in 2021.

She is currently a Lecturer with the College of Information Engineering, China Jiliang University, Hangzhou, China. She served as a Postdoctoral Fellow in computer science with Zhejiang Normal University, Hangzhou, China, and a Visiting Scholar with York University, Toronto, Canada. Her research interests include information retrieval, graph neural networks, data mining, and machine learning.

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